

GRI 305: Emissions 2016

EFFECTIVE DATE: 1 JULY 2018

TOPIC STANDARD

305

GRI 305: Emissions 2016

Topic Standard

Effective Date

This Standard is effective for reports or other materials published on or after 1 July 2018

Responsibility

This Standard is issued by the [Global Sustainability Standards Board \(GSSB\)](#). Any feedback on the GRI Standards can be submitted to gssbsecretariat@globalreporting.org for the consideration of the GSSB.

Due Process

This Standard was developed in the public interest and in accordance with the requirements of the GSSB Due Process Protocol. It has been developed using multi-stakeholder expertise, and with regard to authoritative intergovernmental instruments and widely held expectations of organizations relating to social, environmental, and economic responsibilities.

Legal Liability

This document, designed to promote sustainability reporting, has been developed by the Global Sustainability Standards Board (GSSB) through a unique multi-stakeholder consultative process involving representatives from organizations and report information users from around the world. While the GRI Supervisory Board, Management Board, and GSSB encourage the use of the GRI Sustainability Reporting Standards (GRI Standards) and related Interpretations by all organizations, the preparation and publication of reports based fully or partially on the GRI Standards and related Interpretations are the full responsibility of those producing them. Neither the GRI Supervisory Board, Management Board, GSSB, nor Stichting Global Reporting Initiative (GRI) can assume responsibility for any consequences or damages resulting directly or indirectly from the use of the GRI Standards and related Interpretations in the preparation of reports, or the use of reports based on the GRI Standards and related Interpretations.

Copyright and trademark notice

This document is copyright-protected by Stichting Global Reporting Initiative (GRI). The reproduction and distribution of this document for information and/or use in preparing a sustainability report is permitted without prior permission from GRI. However, neither this document nor any extract from it may be reproduced, stored, translated, or transferred in any form or by any means (electronic, mechanical, photocopied, recorded, or otherwise) for any other purpose without prior written permission from GRI.

Global Reporting Initiative, GRI and logo, GSSB and logo, and GRI Sustainability Reporting Standards (GRI Standards) and logo are trademarks of Stichting Global Reporting Initiative.

© 2025 GRI. All rights reserved.

ISBN 978-90-8866-108-2

Introduction

GRI 305: Emissions 2016 contains disclosures for organizations to report information about their emissions-related impacts, and how they manage these impacts.

The Standard is structured as follows:

- [Section 1](#) contains requirements, which provide information about how the organization manages its emissions-related impacts.
- [Section 2](#) contains seven disclosures, which provide information about the organization's emissions-related impacts.
- The [Glossary](#) contains defined terms with a specific meaning when used in the GRI Standards. The terms are underlined in the text of the GRI Standards and linked to the definitions.
- The [Bibliography](#) lists authoritative intergovernmental instruments and additional references used in developing this Standard.

The rest of the Introduction section provides a background on the topic, an overview of the system of GRI Standards and further information on using this Standard.

Background on the topic

This Standard addresses emissions into air, which are the discharge of substances from a source into the atmosphere. Types of emissions include: greenhouse gas (GHG), ozone-depleting substances (ODS), and nitrogen oxides (NO_x) and sulfur oxides (SO_x), among other significant air emissions.

GHG emissions

GHG emissions are a major contributor to climate change and are governed by the United Nations (UN) 'Framework Convention on Climate Change' and the subsequent UN 'Kyoto Protocol'.

This Standard covers the following GHGs:

- Carbon dioxide (CO₂)
- Methane (CH₄)
- Nitrous oxide (N₂O)
- Hydrofluorocarbons (HFCs)
- Perfluorocarbons (PFCs)
- Sulphur hexafluoride (SF₆)
- Nitrogen trifluoride (NF₃)

Some GHGs, including methane, are also air pollutants that have significant negative impacts on ecosystems, air quality, agriculture, and human and animal health.

As a result, different national and international regulations and incentive systems, such as emissions trading, aim to control the volume and reward the reduction of GHG emissions.

The requirements for GHG emissions in this Standard are based on the requirements of the 'GHG Protocol Corporate Accounting and Reporting Standard' ('GHG Protocol Corporate Standard') and the 'GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard' ('GHG Protocol Corporate Value Chain Standard'). These two standards are part of the GHG Protocol developed by the World Resources Institute (WRI) and the World Business Council on Sustainable Development (WBCSD).

The GHG Protocol has established a classification of GHG emissions called 'Scope': Scope 1, Scope 2 and Scope 3. The GHG emissions standard published by the International Organization for Standardization (ISO), 'ISO 14064', represents these classifications of Scope with the following terms:

- Direct GHG emissions = Scope 1
- Energy indirect GHG emissions = Scope 2
- Other indirect GHG emissions = Scope 3

In this Standard, these terms are combined in the following way, as defined in the [Glossary](#) section:

- Direct (Scope 1) GHG emissions
- Energy indirect (Scope 2) GHG emissions
- Other indirect (Scope 3) GHG emissions

Ozone-depleting substances (ODS)

The ozone layer filters out most of the sun's biologically harmful ultraviolet (UV-B) radiation. Observed and projected ozone depletion due to ODS generates worldwide concern. The UN Environment Programme (UNEP) 'Montreal Protocol on Substances that Deplete the Ozone Layer' ('Montreal Protocol') regulates the phase-out of ODS internationally.

Nitrogen oxides (NO_x), sulfur oxides (SO_x), and other significant air emissions

Pollutants such as NO_x and SO_x have negative impacts on climate, ecosystems, air quality, habitats, agriculture, and human and animal health. Deterioration of air quality, acidification, forest degradation and public health concerns have led to local and international regulations to control emissions of these pollutants.

Reductions in the emission of regulated pollutants lead to improved health conditions for workers and local communities and can enhance relations with affected stakeholders. In regions with emission caps, the volume of emissions also has direct cost implications.

Other significant air emissions include, for example, persistent organic pollutants or particulate matter, as well as air emissions that are regulated under international conventions and/or national laws or regulations, including those listed on an organization's environmental permits.

System of GRI Standards

This Standard is part of the GRI Sustainability Reporting Standards (GRI Standards). The GRI Standards enable an organization to report information about its most significant impacts on the economy, environment, and people, including impacts on their human rights, and how it manages these impacts.

The GRI Standards are structured as a system of interrelated standards that are organized into three series: GRI Universal Standards, GRI Sector Standards, and GRI Topic Standards (see [Figure 1](#) in this Standard).

Universal Standards: GRI 1, GRI 2 and GRI 3

[GRI 1: Foundation 2021](#) specifies the requirements that the organization must comply with to report in accordance with the GRI Standards. The organization begins using the GRI Standards by consulting [GRI 1](#).

[GRI 2: General Disclosures 2021](#) contains disclosures that the organization uses to provide information about its reporting practices and other organizational details, such as its activities, governance, and policies.

[GRI 3: Material Topics 2021](#) provides guidance on how to determine material topics. It also contains disclosures that the organization uses to report information about its process of determining material topics, its list of material topics, and how it manages each topic.

Sector Standards

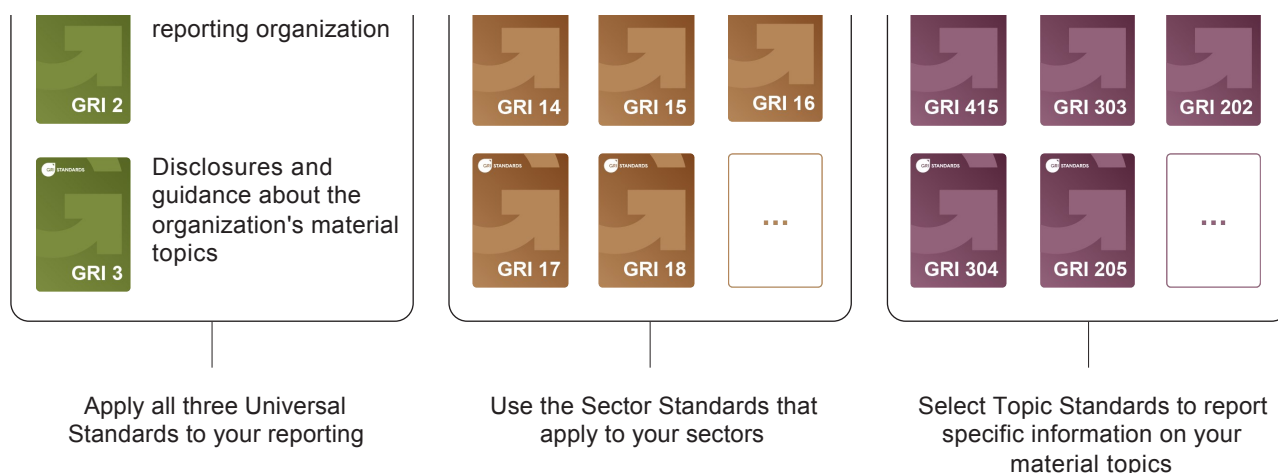
The Sector Standards provide information for organizations about their likely material topics. The organization uses the Sector Standards that apply to its sectors when determining its material topics and when determining what to report for each material topic.

Topic Standards

The Topic Standards contain disclosures that the organization uses to report information about its impacts in relation to particular topics. The organization uses the Topic Standards according to the list of material topics it has determined using [GRI 3](#).

Figure 1. GRI Standards: Universal, Sector and Topic Standards





Using this Standard

This Standard can be used by any organization – regardless of size, type, sector, geographic location, or reporting experience – to report information about its emissions-related impacts.

An organization reporting in accordance with the GRI Standards is required to report the following disclosures if it has determined emissions to be a material topic:

- [Disclosure 3-3 in GRI 3: Material Topics 2021](#) (see clause 1.1 in this Standard);
- [Clause 1.2](#) in this Standard, if it is relevant to its emissions-related impacts;
- Any disclosures from this Topic Standard that are relevant to the organization's emissions-related impacts (Disclosure 305-1 through Disclosure 305-7).

See [Requirements 4 and 5 in GRI 1: Foundation 2021](#).

Reasons for omission are permitted for these requirements and disclosures.

If the organization cannot comply with a disclosure or with a requirement in a disclosure (e.g., because the required information is confidential or subject to legal prohibitions), the organization is required to specify the disclosure or the requirement it cannot comply with, and provide a reason for omission together with an explanation in the GRI content index. See [Requirement 6 in GRI 1: Foundation 2021](#) for more information on reasons for omission.

If the organization cannot report the required information about an item specified in a disclosure because the item (e.g., committee, policy, practice, process) does not exist, it can comply with the requirement by reporting this to be the case. The organization can explain the reasons for not having this item, or describe any plans to develop it. The disclosure does not require the organization to implement the item (e.g., developing a policy), but to report that the item does not exist.

If the organization intends to publish a standalone sustainability report, it does not need to repeat information that it has already reported publicly elsewhere, such as on web pages or in its annual report. In such a case, the organization can report a required disclosure by providing a reference in the GRI content index as to where this information can be found (e.g., by providing a link to the web page or citing the page in the annual report where the information has been published).

Requirements, guidance and defined terms

The following apply throughout this Standard:

Requirements are presented in **bold font** and indicated by the word 'shall'. An organization must comply with requirements to report in accordance with the GRI Standards.

Requirements may be accompanied by guidance.

Guidance includes background information, explanations, and examples to help the organization better understand the requirements. The organization is not required to comply with guidance.

The Standards may also include recommendations. These are cases where a particular course of action is encouraged but not required.

The word 'should' indicates a recommendation, and the word 'can' indicates a possibility or option.

Defined terms are underlined in the text of the GRI Standards and linked to their definitions in the [Glossary](#). The organization is required to apply the definitions in the Glossary.

1. Topic management disclosures

An organization reporting in accordance with the GRI Standards is required to report how it manages each of its material topics.

An organization that has determined emissions to be a material topic is required to report how it manages the topic using [Disclosure 3-3 in GRI 3: Material Topics 2021](#) (see clause 1.1 in this section). The organization is also required to report clause 1.2 in this section, if it is relevant to its emissions-related impacts.

This section is therefore designed to supplement – and not replace – Disclosure 3-3 in *GRI 3*.

2. Topic disclosures

Disclosure 305-1 Direct (Scope 1) GHG emissions

REQUIREMENTS

The reporting organization shall report the following information:

- a. Gross direct (Scope 1) GHG emissions in metric tons of CO₂ equivalent.
- b. Gases included in the calculation; whether CO₂, CH₄, N₂O, HFCs, PFCs, SF₆, NF₃, or all.
- c. Biogenic CO₂ emissions in metric tons of CO₂ equivalent.
- d. Base year for the calculation, if applicable, including:
 - i. the rationale for choosing it;
 - ii. emissions in the base year;
 - iii. the context for any significant changes in emissions that triggered recalculations of base year emissions.
- e. Source of the emission factors and the global warming potential (GWP) rates used, or a reference to the GWP source.
- f. Consolidation approach for emissions; whether equity share, financial control, or operational control.
- g. Standards, methodologies, assumptions, and/or calculation tools used.

Compilation requirements

- 2.1 When compiling the information specified in Disclosure 305-1, the reporting organization shall:
 - 2.1.1 exclude any GHG trades from the calculation of gross direct (Scope 1) GHG emissions;
 - 2.1.2 report biogenic emissions of CO₂ from the combustion or biodegradation of biomass separately from the gross direct (Scope 1) GHG emissions. Exclude biogenic emissions of other types of GHG (such as CH₄ and N₂O), and biogenic emissions of CO₂ that occur in the life cycle of biomass other than from combustion or biodegradation (such as GHG emissions from processing or transporting biomass).

RECOMMENDATIONS

- 2.2 When compiling the information specified in Disclosure 305-1, the reporting organization should:
 - 2.2.1 apply emission factors and GWP rates consistently for the data disclosed;
 - 2.2.2 use the GWP rates from the IPCC assessment reports based on a 100-year timeframe;
 - 2.2.3 select a consistent approach for consolidating direct (Scope 1) and energy indirect (Scope 2) GHG emissions; choosing from the equity share, financial

control, or operational control methods outlined in the 'GHG Protocol Corporate Standard';

2.2.4 if subject to different standards and methodologies, describe the approach to selecting them;

2.2.5 where it aids transparency or comparability over time, provide a breakdown of the direct (Scope 1) GHG emissions by:

2.2.5.1 business unit or facility;

2.2.5.2 country;

2.2.5.3 type of source (stationary combustion, process, fugitive);

2.2.5.4 type of activity.

GUIDANCE

Guidance for Disclosure 305-1

Direct (Scope 1) GHG emissions include, but are not limited to, the CO₂ emissions from the fuel consumption as reported in [Disclosure 302-1 of GRI 302: Energy 2016](#).

Direct (Scope 1) GHG emissions can come from the following sources owned or controlled by an organization:

- Generation of electricity, heating, cooling and steam: these emissions result from combustion of fuels in stationary sources, such as boilers, furnaces, and turbines – and from other combustion processes such as flaring;
- Physical or chemical processing: most of these emissions result from the manufacturing or processing of chemicals and materials, such as cement, steel, aluminum, ammonia, and waste processing;
- Transportation of materials, products, waste, workers, and passengers: these emissions result from the combustion of fuels in mobile combustion sources owned or controlled by the organization, such as trucks, trains, ships, airplanes, buses, and cars;
- Fugitive emissions: these are emissions that are not physically controlled but result from intentional or unintentional releases of GHGs. These can include equipment leaks from joints, seals, packing, and gaskets; methane emissions (e.g., from coal mines) and venting; HFC emissions from refrigeration and air conditioning equipment; and methane leakages (e.g., from gas transport).

Methodologies used to calculate the direct (Scope 1) GHG emissions can include:

- direct measurement of energy source consumed (coal, gas) or losses (refills) of cooling systems and conversion to GHG (CO₂ equivalents);
- mass balance calculations;
- calculations based on site-specific data, such as for fuel composition analysis;
- calculations based on published criteria, such as emission factors and GWP rates;
- direct measurements of GHG emissions, such as continuous online analyzers;
- estimations.

If estimations are used due to a lack of default figures, the reporting organization can indicate the basis and assumptions on which figures were estimated.

For recalculations of prior year emissions, the organization can follow the approach in the 'GHG Protocol Corporate Standard'.

The chosen emission factors can originate from mandatory reporting requirements, voluntary reporting frameworks, or industry groups.

Estimates of GWP rates change over time as scientific research develops. GWP rates from the *Second Assessment Report* of the Intergovernmental Panel on Climate Change (IPCC) are used as the basis for international negotiations under the 'Kyoto Protocol'. Thus, such rates can be used for disclosing GHG emissions where it does not conflict with national or regional reporting requirements. The organization can also use the latest GWP rates from the most recent IPCC assessment report.

The organization can combine Disclosure 305-1 with Disclosures 305-2 (energy indirect/Scope 2 GHG emissions) and 305-3 (other indirect/Scope 3 GHG emissions) to disclose total GHG emissions.

Further details and guidance are available in the 'GHG Protocol Corporate Standard'. See also references [1], [2], [12], [13], [14] and [19] in the [Bibliography](#).

Disclosure 305-2 Energy indirect (Scope 2) GHG emissions

REQUIREMENTS

The reporting organization shall report the following information:

- a. Gross location-based energy indirect (Scope 2) GHG emissions in metric tons of CO₂ equivalent.
- b. If applicable, gross market-based energy indirect (Scope 2) GHG emissions in metric tons of CO₂ equivalent.
- c. If available, the gases included in the calculation; whether CO₂, CH₄, N₂O, HFCs, PFCs, SF₆, NF₃, or all.
- d. Base year for the calculation, if applicable, including:
 - i. the rationale for choosing it;
 - ii. emissions in the base year;
 - iii. the context for any significant changes in emissions that triggered recalculations of base year emissions.
- e. Source of the emission factors and the global warming potential (GWP) rates used, or a reference to the GWP source.
- f. Consolidation approach for emissions; whether equity share, financial control, or operational control.
- g. Standards, methodologies, assumptions, and/or calculation tools used.

Compilation requirements

- 2.3 When compiling the information specified in Disclosure 305-2, the reporting organization shall:
 - 2.3.1 exclude any GHG trades from the calculation of gross energy indirect (Scope 2) GHG emissions;
 - 2.3.2 exclude other indirect (Scope 3) GHG emissions that are disclosed as specified in Disclosure 305-3;
 - 2.3.3 account and report energy indirect (Scope 2) GHG emissions based on the location-based method, if it has operations in markets without product or supplier-specific data;
 - 2.3.4 account and report energy indirect (Scope 2) GHG emissions based on both the location-based and market-based methods, if it has any operations in markets providing product or supplier-specific data in the form of contractual instruments.

RECOMMENDATIONS

- 2.4 When compiling the information specified in Disclosure 305-2, the reporting organization should:
 - 2.4.1 apply emission factors and GWP rates consistently for the data disclosed;
 - 2.4.2 use the GWP rates from the IPCC assessment reports based on a 100-year timeframe;
 - 2.4.3 select a consistent approach for consolidating direct (Scope 1) and energy indirect (Scope 2) GHG emissions, choosing from the equity share, financial control, or operational control methods outlined in the 'GHG Protocol Corporate Standard';
 - 2.4.4 if subject to different standards and methodologies, describe the approach to selecting them;
 - 2.4.5 where it aids transparency or comparability over time, provide a breakdown of the energy indirect (Scope 2) GHG emissions by:
 - 2.4.5.1 business unit or facility;
 - 2.4.5.2 country;
 - 2.4.5.3 type of source (electricity, heating, cooling, and steam);
 - 2.4.5.4 type of activity.

GUIDANCE

Guidance for Disclosure 305-2

Energy indirect (Scope 2) GHG emissions include, but are not limited to, the CO₂ emissions from the generation of purchased or acquired electricity, heating, cooling, and steam consumed by an organization – disclosed as specified in [Disclosure 302-1 of GRI 302: Energy 2016](#). For many organizations, the energy indirect (Scope 2) GHG emissions that result from the generation of purchased electricity can be much greater than their direct (Scope 1) GHG emissions.

The 'GHG Protocol Scope 2 Guidance' requires organizations to provide two distinct Scope 2 values: a location-based and a market-based value. A location-based method reflects the average GHG emissions intensity of grids on which energy consumption occurs, using mostly grid-average emission factor data. A market-based method reflects emissions from electricity that an organization has purposefully chosen (or its lack of choice). It derives emission factors from contractual instruments, which include any type of contract between two parties for the sale and purchase of energy bundled with attributes about the energy generation, or for unbundled attribute claims.

The market-based method calculation also includes the use of a residual mix, if the organization does not have specified emissions-intensity from its contractual instruments. This helps prevent double counting between consumers' market-based method figures. If a residual mix is unavailable, the organization can disclose this and use grid-average emission factors as a proxy (which can mean that the location-based and market-based are the same number until information on the residual mix is available).

The reporting organization can apply the Quality Criteria in the 'GHG Protocol Scope 2 Guidance' so that contractual instruments convey GHG emission rate claims and to prevent double counting. See reference [18] in the [Bibliography](#).

For recalculations of prior year emissions, the organization can follow the approach in the 'GHG Protocol Corporate Standard'.

The chosen emission factors can originate from mandatory reporting requirements, voluntary reporting frameworks, or industry groups.

Estimates of GWP rates change over time as scientific research develops. GWP rates from the *Second Assessment Report* of the IPCC are used as the basis for international negotiations under the 'Kyoto Protocol'. Thus, such rates can be used for disclosing GHG emissions where it does not conflict with national or regional reporting requirements. The organization can also use the latest GWP rates from the most recent IPCC assessment report.

The organization can combine Disclosure 305-2 with Disclosures 305-1 (direct/Scope 1 GHG emissions) and 305-3 (other indirect/Scope 3 GHG emissions) to disclose total GHG emissions.

Further details and guidance are available in the 'GHG Protocol Corporate Standard'. Details on the location-based and market-based methods are available in the 'GHG Protocol Scope 2 Guidance'. See also references [1], [2], [12], [13], [14] and [18] in the [Bibliography](#).

Disclosure 305-3 Other indirect (Scope 3) GHG emissions

REQUIREMENTS

The reporting organization shall report the following information:

- a. **Gross other indirect (Scope 3) GHG emissions in metric tons of CO₂ equivalent.**
- b. **If available, the gases included in the calculation; whether CO₂, CH₄, N₂O, HFCs, PFCs, SF₆, NF₃, or all.**
- c. **Biogenic CO₂ emissions in metric tons of CO₂ equivalent.**
- d. **Other indirect (Scope 3) GHG emissions categories and activities included in the calculation.**
- e. **Base year for the calculation, if applicable, including:**

- i. the rationale for choosing it;
 - ii. emissions in the base year;
 - iii. the context for any significant changes in emissions that triggered recalculations of base year emissions.
- f. Source of the emission factors and the global warming potential (GWP) rates used, or a reference to the GWP source.
- g. Standards, methodologies, assumptions, and/or calculation tools used.

Compilation requirements

- 2.5 When compiling the information specified in Disclosure 305-3, the reporting organization shall:

- 2.5.1 exclude any GHG trades from the calculation of gross other indirect (Scope 3) GHG emissions;
- 2.5.2 exclude energy indirect (Scope 2) GHG emissions from this disclosure. Energy indirect (Scope 2) GHG emissions are disclosed as specified in Disclosure 305-2;
- 2.5.3 report biogenic emissions of CO₂ from the combustion or biodegradation of biomass that occur in its value chain separately from the gross other indirect (Scope 3) GHG emissions. Exclude biogenic emissions of other types of GHG (such as CH₄ and N₂O), and biogenic emissions of CO₂ that occur in the life cycle of biomass other than from combustion or biodegradation (such as GHG emissions from processing or transporting biomass).

RECOMMENDATIONS

- 2.6 When compiling the information specified in Disclosure 305-3, the reporting organization should:
- 2.6.1 apply emission factors and GWP rates consistently for the data disclosed;
 - 2.6.2 use the GWP rates from the IPCC assessment reports based on a 100-year timeframe;
 - 2.6.3 if subject to different standards and methodologies, describe the approach to selecting them;
 - 2.6.4 list other indirect (Scope 3) GHG emissions, with a breakdown by upstream and downstream categories and activities;
 - 2.6.5 where it aids transparency or comparability over time, provide a breakdown of the other indirect (Scope 3) GHG emissions by:
 - 2.6.5.1 business unit or facility;
 - 2.6.5.2 country;
 - 2.6.5.3 type of source;
 - 2.6.5.4 type of activity.

GUIDANCE

Guidance for Disclosure 305-3

Other indirect (Scope 3) GHG emissions are a consequence of an organization's activities, but occur from sources not owned or controlled by the organization. Other indirect (Scope 3) GHG emissions include both upstream and downstream emissions. Some examples of Scope 3 activities include extracting and producing purchased materials; transporting purchased fuels in vehicles not owned or controlled by the organization; and the end use of products and services.

Other indirect emissions can also come from the decomposing of the organization's waste. Process-related emissions during the manufacture of purchased goods and fugitive emissions in facilities not owned by the organization can also produce indirect emissions.

For some organizations, GHG emissions that result from energy consumption outside of the organization can be much greater than their direct (Scope 1) or energy indirect (Scope 2) GHG emissions.

The reporting organization can identify other indirect (Scope 3) GHG emissions by assessing which of its activities' emissions:

- contribute significantly to the organization's total anticipated other indirect (Scope 3) GHG emissions;

- offer potential for reductions the organization can undertake or influence;
- contribute to climate change-related risks, such as financial, regulatory, supply chain, product and customer, litigation, and reputational risks;
- are deemed material by stakeholders, such as customers, suppliers, investors, or civil society;
- result from outsourced activities previously performed in-house, or that are typically performed in-house by other organizations in the same sector;
- have been identified as significant for the organization's sector;
- meet any additional criteria for determining relevance, developed by the organization or by organizations in its sector.

The organization can use the following upstream and downstream categories and activities from the 'GHG Protocol Corporate Value Chain Standard' (see reference [15] in the [Bibliography](#)).

Upstream categories

1. Purchased goods and services
2. Capital goods
3. Fuel- and energy-related activities (not included in Scope 1 or Scope 2)
4. Upstream transportation and distribution
5. Waste generated in operations
6. Business travel
7. Employee commuting
8. Upstream leased assets
- Other upstream

Downstream categories

9. Downstream transportation and distribution
10. Processing of sold products
11. Use of sold products
12. End-of-life treatment of sold products
13. Downstream leased assets
14. Franchises
15. Investments
- Other downstream

For each of these categories and activities, the organization can provide a figure in CO₂ equivalent or explain why certain data are not included.

For recalculations of prior year emissions, the organization can follow the approach in the 'GHG Protocol Corporate Value Chain Standard'.

The chosen emission factors can originate from mandatory reporting requirements, voluntary reporting frameworks, or industry groups.

Estimates of GWP rates change over time as scientific research develops. GWP rates from the *Second Assessment Report* of the IPCC are used as the basis for international negotiations under the 'Kyoto Protocol'. Thus, such rates can be used for disclosing GHG emissions where it does not conflict with national or regional reporting requirements. The organization can also use the latest GWP rates from the most recent IPCC assessment report.

The organization can combine Disclosure 305-3 with Disclosures 305-1 (direct/Scope 1 GHG emissions) and 305-2 (energy indirect/Scope 2 GHG emissions) to disclose total GHG emissions.

See references [1], [2], [12], [13], [15], [17] and [19] in the [Bibliography](#).

Disclosure 305-4 GHG emissions intensity

REQUIREMENTS

The reporting organization shall report the following information:

- a. **GHG emissions intensity ratio for the organization.**
- b. **Organization-specific metric (the denominator) chosen to calculate the ratio.**
- c. **Types of GHG emissions included in the intensity ratio; whether direct (Scope 1), energy indirect (Scope 2), and/or other indirect (Scope 3).**
- d. **Gases included in the calculation; whether CO₂, CH₄, N₂O, HFCs, PFCs, SF₆, NF₃, or all.**

Compilation requirements

- 2.7 **When compiling the information specified in Disclosure 305-4, the reporting organization shall:**
 - 2.7.1 **calculate the ratio by dividing the absolute GHG emissions (the numerator) by the organization-specific metric (the denominator);**
 - 2.7.2 **if reporting an intensity ratio for other indirect (Scope 3) GHG emissions, report this intensity ratio separately from the intensity ratios for direct (Scope 1) and energy indirect (Scope 2) emissions.**

RECOMMENDATIONS

- 2.8 When compiling the information specified in Disclosure 305-4, the reporting organization should, where it aids transparency or comparability over time, provide a breakdown of the GHG emissions intensity ratio by:
 - 2.8.1 business unit or facility;
 - 2.8.2 country;
 - 2.8.3 type of source;
 - 2.8.4 type of activity.

GUIDANCE

Guidance for Disclosure 305-4

Intensity ratios can be provided for, among others:

- products (such as metric tons of CO₂ emissions per unit produced);
- services (such as metric tons of CO₂ emissions per function or per service);
- sales (such as metric tons of CO₂ emissions per sales).

Organization-specific metrics (denominators) can include:

- units of product;
- production volume (such as metric tons, liters, or MWh);
- size (such as m² floor space);
- number of full-time employees;
- monetary units (such as revenue or sales).

The reporting organization can report an intensity ratio for direct (Scope 1) and energy indirect (Scope 2) GHG emissions combined, using the figures reported in Disclosures 305-1 and 305-2.

Background

Intensity ratios define GHG emissions in the context of an organization-specific metric. Many organizations track environmental performance with intensity ratios, which are often called normalized environmental impact data.

GHG emissions intensity expresses the amount of GHG emissions per unit of activity, output, or any other organization-specific metric. In combination with an organization's absolute GHG emissions, reported in Disclosures 305-1, 305-2, and 305-3, GHG emissions intensity helps to contextualize the organization's efficiency, including in relation to other organizations.

See references [13], [14], and [19] in the [Bibliography](#).

Disclosure 305-5 Reduction of GHG emissions

REQUIREMENTS

The reporting organization shall report the following information:

- a. **GHG emissions reduced as a direct result of reduction initiatives, in metric tons of CO₂**

equivalent.

- b. **Gases included in the calculation; whether CO₂, CH₄, N₂O, HFCs, PFCs, SF₆, NF₃, or all.**
- c. **Base year or baseline, including the rationale for choosing it.**
- d. **Scopes in which reductions took place; whether direct (Scope 1), energy indirect (Scope 2), and/or other indirect (Scope 3).**
- e. **Standards, methodologies, assumptions, and/or calculation tools used.**

Compilation requirements

- 2.9 When compiling the information specified in Disclosure 305-5, the reporting organization shall:**

- 2.9.1 exclude reductions resulting from reduced production capacity or outsourcing;**
- 2.9.2 use the inventory or project method to account for reductions;**
- 2.9.3 calculate an initiative's total reductions of GHG emissions as the sum of its associated primary effects and any significant secondary effects;**
- 2.9.4 if reporting two or more Scope types, report the reductions for each separately;**
- 2.9.5 report reductions from offsets separately.**

RECOMMENDATIONS

- 2.10 When compiling the information specified in Disclosure 305-5, the reporting organization should, if subject to different standards and methodologies, describe the approach to selecting them.**

GUIDANCE

Guidance for Disclosure 305-5

The reporting organization can prioritize disclosing reduction initiatives that were implemented in the reporting period, and that have the potential to contribute significantly to reductions. The organization can describe reduction initiatives and their targets when reporting how it manages this topic.

Reduction initiatives can include:

- process redesign;
- conversion and retrofitting of equipment;
- fuel switching;
- changes in behavior;
- offsets.

The organization can report reductions disaggregated by initiatives or groups of initiatives.

This disclosure can be used in combination with Disclosures 305-1, 305-2, and 305-3 of this Standard to monitor the reduction of GHG emissions with reference to the organization's targets, or to regulations and trading systems at international or national level.

See references [12], [13], [14], [15], [16], and [19] in the [Bibliography](#).

Guidance for clause 2.9.2

The inventory method compares reductions to a base year. The project method compares reductions to a baseline. Further details on these methods are available in references [15] and [16] in the [Bibliography](#).

Guidance for clause 2.9.3

Primary effects are the elements or activities designed to reduce GHG emissions, such as carbon storage. Secondary effects are smaller, unintended consequences of a reduction initiative, including changes to production or manufacture, which result in changes to GHG emissions elsewhere. See reference [14] in the [Bibliography](#).

Disclosure 305-6 Emissions of ozone-depleting substances (ODS)

REQUIREMENTS

The reporting organization shall report the following information:

- a. **Production, imports, and exports of ODS in metric tons of CFC-11 (trichlorofluoromethane) equivalent.**
- b. **Substances included in the calculation.**
- c. **Source of the emission factors used.**
- d. **Standards, methodologies, assumptions, and/or calculation tools used.**

Compilation requirements

- 2.11 When compiling the information specified in Disclosure 305-6, the reporting organization shall:**

- 2.11.1 calculate the production of ODS as the amount of ODS produced, minus the amount destroyed by approved technologies, and minus the amount entirely used as feedstock in the manufacture of other chemicals;**

Production of ODS
=
ODS produced
-
ODS destroyed by approved technologies
-
ODS entirely used as feedstock in the manufacture of other chemicals

- 2.11.2 exclude ODS recycled and reused.**

RECOMMENDATIONS

- 2.12 When compiling the information specified in Disclosure 305-6, the reporting organization should:**

- 2.12.1 if subject to different standards and methodologies, describe the approach to selecting them;
- 2.12.2 where it aids transparency or comparability over time, provide a breakdown of the ODS data by:
 - 2.12.2.1 business unit or facility;
 - 2.12.2.2 country;
 - 2.12.2.3 type of source;
 - 2.12.2.4 type of activity.

GUIDANCE

Guidance for Disclosure 305-6

The reporting organization can report separate or combined data for the substances included in the calculation.

Background

Measuring ODS production, imports, and exports helps to indicate how an organization complies with legislation. This is particularly relevant if the organization produces or uses ODS in its processes, products and services and is subject to phase-out commitments. Results on ODS phase-out help to indicate the organization's position in any markets affected by regulation on ODS.

This disclosure covers the substances included in Annexes A, B, C, and E of the 'Montreal Protocol' as well as any other ODS produced, imported, or exported by an organization.

See references [1], [2], [8] and [9] in the [Bibliography](#).

Disclosure 305-7 Nitrogen oxides (NO_x), sulfur oxides (SO_x), and other significant air emissions

REQUIREMENTS

The reporting organization shall report the following information:

- a. **Significant air emissions**, in kilograms or multiples, for each of the following:
 - i. NO_x
 - ii. SO_x
 - iii. Persistent organic pollutants (POP)
 - iv. Volatile organic compounds (VOC)
 - v. Hazardous air pollutants (HAP)
 - vi. Particulate matter (PM)
 - vii. Other standard categories of air emissions identified in relevant regulations
- b. Source of the emission factors used.
- c. Standards, methodologies, assumptions, and/or calculation tools used.

Compilation requirements

- 2.13 When compiling the information specified in Disclosure 305-7, the reporting organization shall select one of the following approaches for calculating significant air emissions:

- 2.13.1 Direct measurement of emissions (such as online analyzers);
- 2.13.2 Calculation based on site-specific data;
- 2.13.3 Calculation based on published emission factors;
- 2.13.4 Estimation. If estimations are used due to a lack of default figures, the organization shall indicate the basis on which figures were estimated.

RECOMMENDATIONS

- 2.14 When compiling the information specified in Disclosure 305-7, the reporting organization should:
 - 2.14.1 if subject to different standards and methodologies, describe the approach to selecting them;
 - 2.14.2 where it aids transparency or comparability over time, provide a breakdown of the air emissions data by:
 - 2.14.2.1 business unit or facility;
 - 2.14.2.2 country;
 - 2.14.2.3 type of source;
 - 2.14.2.4 type of activity.

GUIDANCE

See references [3], [4], [5], [6] and [10] in the [Bibliography](#).

Glossary

This glossary provides definitions for terms used in this Standard. The organization is required to apply these definitions when using the GRI Standards.

The definitions included in this glossary may contain terms that are further defined in the complete [GRI Standards Glossary](#). All defined terms are underlined. If a term is not defined in this glossary or in the complete [GRI Standards Glossary](#), definitions that are commonly used and understood apply.

base year *

historical datum (such as year) against which a measurement is tracked over time

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

baseline

starting point used for comparisons

Note: In the context of energy reporting, the baseline is the projected energy consumption in the absence of any reduction activity.

biogenic carbon dioxide (CO₂) emission

emission of CO₂ from the combustion or biodegradation of biomass

carbon dioxide (CO₂) equivalent *

measure used to compare the emissions from various types of greenhouse gas (GHG) based on their global warming potential (GWP)

Note: The CO₂ equivalent for a gas is determined by multiplying the metric tons of the gas by the associated GWP.

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

CFC11 (trichlorofluoromethane) equivalent

measure used to compare various substances based on their relative ozone depletion potential (ODP)

Note: The reference level of 1 is the potential of CFC-11 (trichlorofluoromethane) and CFC-12 (dichlorodifluoromethane) to cause ozone depletion.

direct (Scope 1) GHG emissions *

greenhouse gas (GHG) emissions from sources that are owned or controlled by the organization

Examples: CO₂ emissions from fuel consumption

Note: A GHG source is any physical unit or process that releases GHG into the atmosphere.

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

energy indirect (Scope 2) GHG emissions *

greenhouse gas (GHG) emissions that result from the generation of purchased or acquired electricity, heating, cooling, and steam consumed by the organization

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

global warming potential (GWP) *

value describing the radiative forcing impact of one unit of a given greenhouse gas (GHG) relative to one unit of CO₂ over a given period of time

Note: GWP values convert GHG emissions data for non-CO₂ gases into units of CO₂ equivalent.

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

greenhouse gas (GHG) *

gas that contributes to the greenhouse effect by absorbing infrared radiation

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

greenhouse gas (GHG) trade *

purchase, sale or transfer of greenhouse gas (GHG) emission offsets or allowances

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

human rights

rights inherent to all human beings, which include, at a minimum, the rights set out in the *United Nations (UN) International Bill of Human Rights* and the principles concerning fundamental rights set out in the *International Labour Organization (ILO) Declaration on Fundamental Principles and Rights at Work*

Source: United Nations (UN), *Guiding Principles on Business and Human Rights: Implementing the United Nations “Protect, Respect and Remedy” Framework*, 2011; modified

Note: See [Guidance to 2-23-b-i in GRI 2: General Disclosures 2021](#) for more information on ‘human rights’.

impact

effect the organization has or could have on the economy, environment, and people, including on their human rights, which in turn can indicate its contribution (negative or positive) to sustainable development

Note 1: Impacts can be actual or potential, negative or positive, short-term or long-term, intended or unintended, and reversible or irreversible.

Note 2: See [section 2.1 in GRI 1: Foundation 2021](#) for more information on ‘impact’.

material topics

topics that represent the organization’s most significant impacts on the economy, environment, and people, including impacts on their human rights

Note: See [section 2.2 in GRI 1: Foundation 2021](#) and [section 1 in GRI 3: Material Topics 2021](#) for more information on ‘material topics’.

other indirect (Scope 3) GHG emissions *

indirect greenhouse gas (GHG) emissions not included in energy indirect (Scope 2) GHG emissions that occur outside of the organization, including both upstream and downstream emissions

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

ozone-depleting substance (ODS)

substance with an ozone depletion potential (ODP) greater than 0 that can deplete the stratospheric ozone layer

Note: Most ODS are controlled under the United Nations Environment Programme (UNEP), *Montreal Protocol on Substances that Deplete the Ozone Layer*, 1987, and its amendments, and include chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), halons, and methyl bromide.

reduction of greenhouse gas (GHG) emissions *

decrease in greenhouse gas (GHG) emissions or increase in removal or storage of GHG from the atmosphere, relative to baseline emissions

Note: Primary effects will result in GHG reductions, as will some secondary effects. An initiative’s total GHG reductions are quantified as the sum of its associated primary effect(s) and any significant secondary effects (which may involve decreases or countervailing increases in GHG emissions).

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated*

term.

scope of GHG emissions *

classification of the operational boundaries where greenhouse gas (GHG) emissions occur

Note 1: Scope classifies whether GHG emissions are created by the organization itself, or are created by other related organizations, for example electricity suppliers or logistics companies.

Note 2: There are three classifications of Scope: Scope 1, Scope 2 and Scope 3.

Note 3: The classification of Scope derives from the World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), *GHG Protocol Corporate Accounting and Reporting Standard*, Revised Edition, 2004.

** Please note this term will be updated following the effective date of GRI 102: Climate Change 2025 and GRI 103: Energy 2025, as of 1 January 2027. Please see GRI 102/103 for the updated term.*

significant air emission

air emission regulated under international conventions and/or national laws or regulations

Note: Significant air emissions include those listed on environmental permits for the organization's operations.

sustainable development / sustainability

development that meets the needs of the present without compromising the ability of future generations to meet their own needs

Source: World Commission on Environment and Development, *Our Common Future*, 1987

Note: The terms 'sustainability' and 'sustainable development' are used interchangeably in the GRI Standards.

Bibliography

This section lists authoritative intergovernmental instruments and additional references used in developing this Standard.

Authoritative instruments:

1. Intergovernmental Panel on Climate Change (IPCC), *Climate Change 1995: The Science of Climate Change, Contribution of Working Group I to the Second Assessment Report of the Intergovernmental Panel on Climate Change*, 1995.
2. Intergovernmental Panel on Climate Change (IPCC), *Climate Change 2007: The Physical Science Basis, Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*, 2007.
3. United Nations Economic Commission for Europe (UNECE) Convention, 'Geneva Protocol concerning the Control of Emissions of Volatile Organic Compounds or their Transboundary Fluxes', 1991.
4. United Nations Economic Commission for Europe (UNECE) Convention, 'Gothenburg Protocol to Abate Acidification, Eutrophication and Ground-level Ozone', 1999.
5. United Nations Economic Commission for Europe (UNECE) Convention, 'Helsinki Protocol on the Reduction of Sulphur Emissions or their Transboundary Fluxes', 1985.
6. United Nations Economic Commission for Europe (UNECE) Convention, 'Sofia Protocol concerning the Control of Emissions of Nitrogen Oxides or their Transboundary Fluxes', 1988.
7. United Nations Environment Programme (UNEP) and World Meteorological Organization (WMO), *Integrated Assessment of Black Carbon and Tropospheric Ozone*, 2011.
8. United Nations Environment Programme (UNEP), 'Montreal Protocol on Substances that Deplete the Ozone Layer', 1987.
9. United Nations Environment Programme (UNEP), *Standards and Codes of Practice to Eliminate Dependency on Halons - Handbook of Good Practices in the Halon Sector*, 2001.

10. United Nations Environment Programme (UNEP) Convention, 'Stockholm Convention on Persistent Organic Pollutants (POPs)', Annex A, B, and C, 2009.
11. United Nations (UN) Framework Convention, 'United Nations Framework Convention on Climate Change', 1992.
12. United Nations (UN) Protocol, 'Kyoto Protocol to the United Nations Framework Convention on Climate Change', 1997.

Additional references:

13. CDP, *Investor CDP Information Request*, updated annually.
14. World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), 'GHG Protocol Corporate Accounting and Reporting Standard', Revised Edition, 2004.
15. World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), 'GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard', 2011.
16. World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), 'GHG Protocol for Project Accounting', 2005.
17. World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), 'GHG Protocol Product Life Cycle Accounting and Reporting Standard', 2011.
18. World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), 'GHG Protocol Scope 2 Guidance. An amendment to the GHG Protocol Corporate Standard', 2015.
19. World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD), 'Greenhouse Gas Protocol Accounting Notes, No. 1, Accounting and Reporting Standard Amendment', 2012.



PO Box 10039
1001 EA Amsterdam
The Netherlands

www.globalreporting.org